Approximating fixed size quantum correlations in polynomial time

Julius A. Zeiss[1], Gereon Koßmann[1], Omar Fawzi[2] and Mario Berta[1]

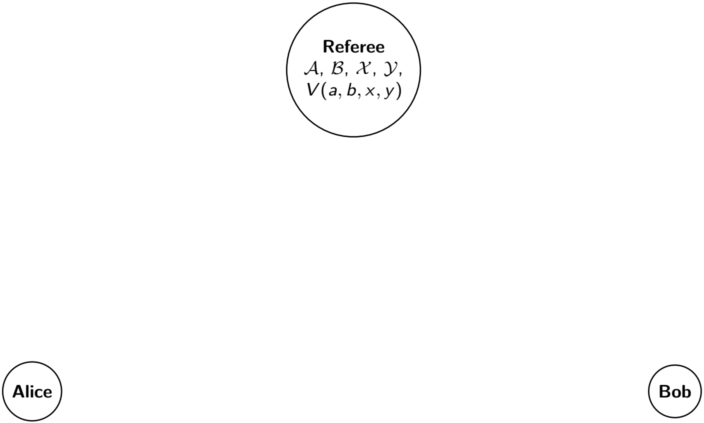
[1] *Institute for Quantum Information, RWTH Aachen University, Aachen, Germany*
[2] *Univ Lyon, Inria, ENS Lyon, UCBL, LIP, Lyon, France*

15.04.2026, Quantum Information Seminar, Cologne, Germany

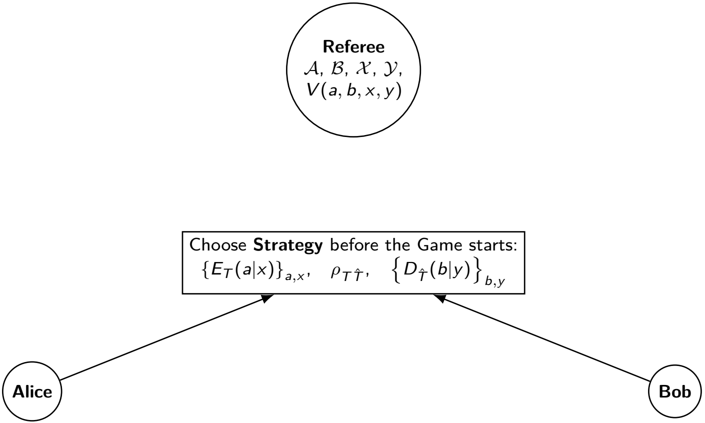
AQEC 2507.12326 , **Non-local Games** 2507.12302, **GPTs** 2601.15184

Non-local Games

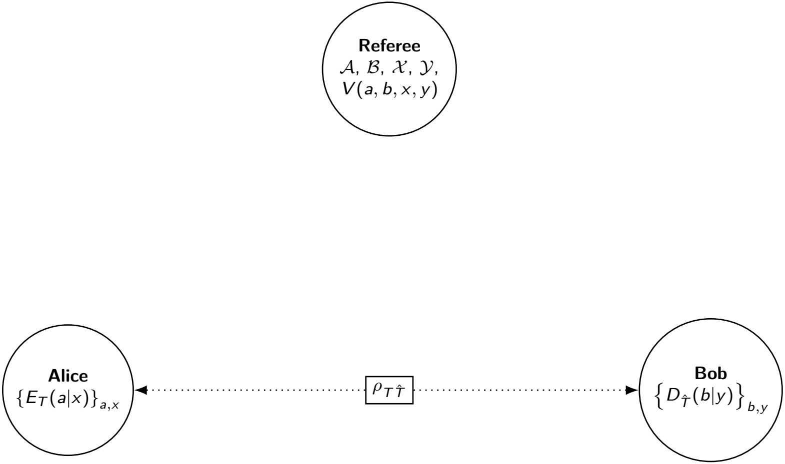
Free and fixed-sized non-local games

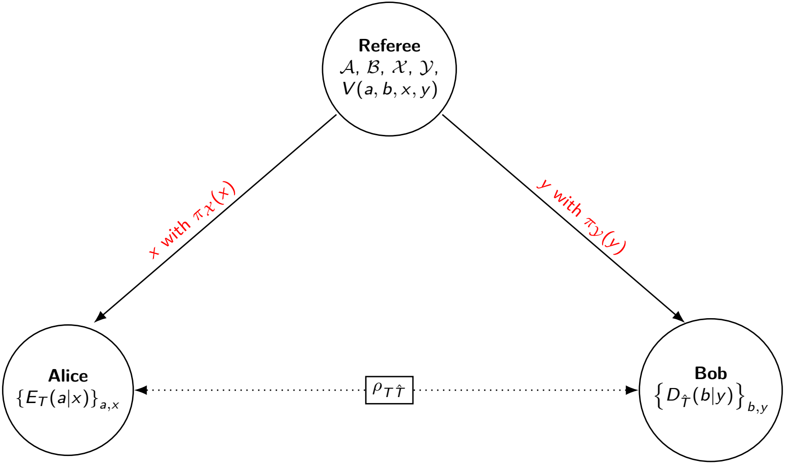


Free and fixed-sized non-local games

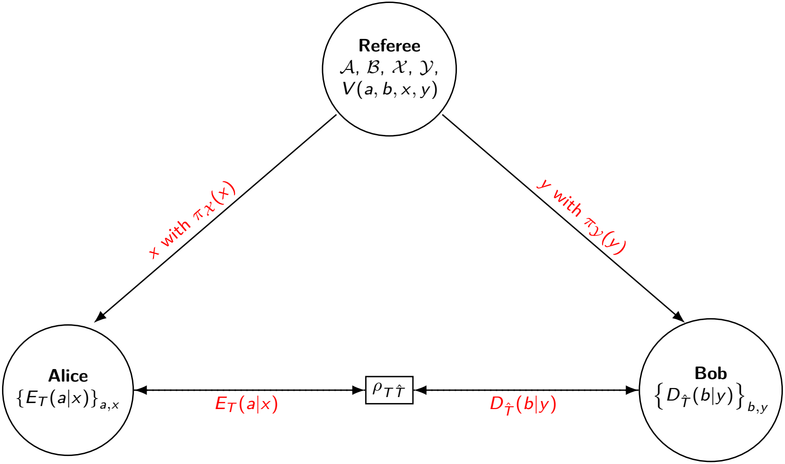


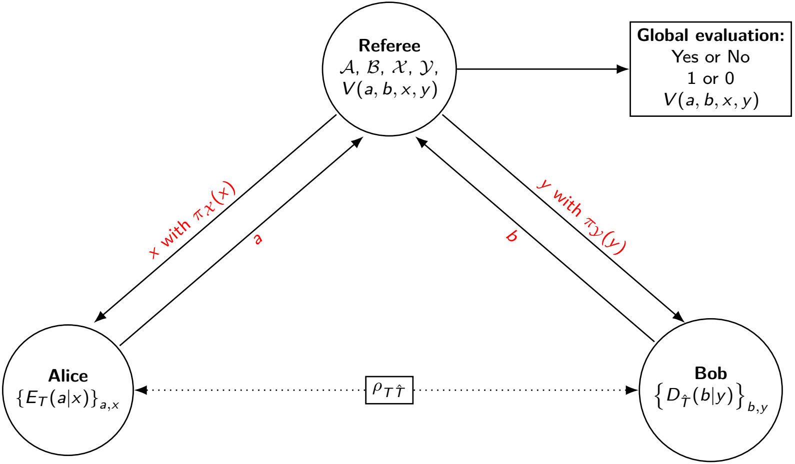
Free and fixed-sized non-local games




Referee
$\mathcal{A}$, $\mathcal{B}$, $\mathcal{X}$, $\mathcal{Y}$,
$V(a, b, x, y)$
x with $\pi_{\mathcal{X}}(x)$
y with $\pi_{\mathcal{Y}}(y)$
Alice
$\{E_T(a|x)\}_{a,x}$
$\rho_{T\hat{T}}$
Bob
$\left\{D_{\hat{T}}(b|y)\right\}_{b,y}$

Free and fixed-sized non-local games




Referee
$\mathcal{A}$, $\mathcal{B}$, $\mathcal{X}$, $\mathcal{Y}$,
$V(a, b, x, y)$
Global evaluation:
Yes or No
1 or 0
$V(a, b, x, y)$
x with $\pi_{\mathcal{X}}(x)$
a
y with $\pi_{\mathcal{Y}}(y)$
b
Alice
$\{E_T(a|x)\}_{a,x}$
$\rho_{T\hat{T}}$
Bob
$\left\{D_{\hat{T}}(b|y)\right\}_{b,y}$

- $\mathbf{T = 1}$: Classical strategy [**kempe2010unique**], PCP theorem [**arora1998proof**; **arora1998probabilistic**]: **NP-hard** to approximate game value up to constant multiplicative factor

- $\mathbf{T = 1}$: Classical strategy [**kempe2010unique**], PCP theorem [**arora1998proof**; **arora1998probabilistic**]: **NP-hard** to approximate game value up to constant multiplicative factor
- $\mathbf{T} \to \infty$: unbounded games [**cleve2004consequences**; **ito2009oracularization**; **kempe2011entangled**; **brandao˙harrow˙2014**] Computing game value even **harder**: Approximations cannot be done in finite time [**ji2021mip**], Further challenges: tensor product vs. commuting subalgebras

Hardness & the bounded-entanglement regime

- $\mathbf{T} = \mathbf{1}$: Classical strategy [**kempe2010unique**], PCP theorem [**arora1998proof**; **arora1998probabilistic**]: **NP-hard** to approximate game value up to constant multiplicative factor
- $\mathbf{T} \to \infty$: unbounded games [**cleve2004consequences**; **ito2009oracularization**; **kempe2011entangled**; **brandao˙harrow˙2014**] Computing game value even **harder**: Approximations cannot be done in finite time [**ji2021mip**], Further challenges: tensor product vs. commuting subalgebras
- $\mathbf{1} < \mathbf{T} < \infty$: [**navascues2014characterization**; **navascues2015characterizing**; **navascues2015bounding**; **jee2020quasi**], **Operationally relevant and tractable**: Self-testing and Pseudo-telepathy games...

Non-local games as optimization problems

$$w_{Q(T)}(V,\pi) = |T| \max_{(\rho,\,E,\,D)} \quad \mathrm{Tr}\left[\left(V_{\mathcal{ABXY}} \otimes \rho_{T\hat{T}}\right)\left(E_{\mathcal{AX}T} \otimes D_{\mathcal{BY}\hat{T}}\right)\right]$$

s.t.

$$\rho_{T\hat{T}} \succcurlyeq 0, \qquad \mathrm{Tr}\left[\rho_{T\hat{T}}\right] = 1,$$

$$E_{\mathcal{AX}T} = \sum_{a,x} \pi_{\mathcal{X}}(x) \left|ax\right\rangle \left\langle ax\right|_{\mathcal{AX}} \otimes E_T(a|x) \succcurlyeq 0,$$

$$E_{\mathcal{X}T} = \sum_{x} \pi_{\mathcal{X}}(x) \left|x\right\rangle \left\langle x\right|_{\mathcal{X}} \otimes \mathcal{I}_T,$$

$$D_{A_2 Q_2 \hat{T}} = \sum_{b,y} \pi_{\mathcal{Y}}(b) \left|by\right\rangle \left\langle by\right|_{\mathcal{BY}} \otimes \frac{D_{\hat{T}}(b|y)}{|\hat{T}|} \succcurlyeq 0,$$

$$D_{\mathcal{Y}\hat{T}} = \sum_{y} \pi_{\mathcal{Y}}(y) \left|y\right\rangle \left\langle y\right|_{\mathcal{Y}} \otimes \frac{\mathcal{I}_{\hat{T}}}{|\hat{T}|},$$

Non-local games as optimization problems

$$
\begin{aligned}
w_{Q(T)}(V,\pi) = |T| \max_{(\rho,\,E,\,D)} \quad & \mathrm{Tr}\left[\left(V_{\mathcal{A}\mathcal{B}\mathcal{X}\mathcal{Y}} \otimes \rho_{T\hat{T}}\right)\left(E_{\mathcal{A}\mathcal{X}T} \otimes D_{\mathcal{B}\mathcal{Y}\hat{T}}\right)\right] \\
\text{s.t.} \quad & \rho_{T\hat{T}} \succcurlyeq 0, \qquad \mathrm{Tr}\left[\rho_{T\hat{T}}\right] = 1, \\
& E_{\mathcal{A}\mathcal{X}T} = \sum_{a,x} \pi_{\mathcal{X}}(x) \left|ax\right\rangle\left\langle ax\right|_{\mathcal{A}\mathcal{X}} \otimes E_T(a|x) \succcurlyeq 0, \\
& E_{\mathcal{X}T} = \sum_{x} \pi_{\mathcal{X}}(x) \left|x\right\rangle\left\langle x\right|_{\mathcal{X}} \otimes \mathcal{I}_T, \\
& D_{A_2Q_2\hat{T}} = \sum_{b,y} \pi_{\mathcal{Y}}(b) \left|by\right\rangle\left\langle by\right|_{\mathcal{B}\mathcal{Y}} \otimes \frac{D_{\hat{T}}(b|y)}{|\hat{T}|} \succcurlyeq 0, \\
& D_{\mathcal{Y}\hat{T}} = \sum_{y} \pi_{\mathcal{Y}}(y) \left|y\right\rangle\left\langle y\right|_{\mathcal{Y}} \otimes \frac{\mathcal{I}_{\hat{T}}}{|\hat{T}|},
\end{aligned}
$$

Problem

How well and at what computational cost can we approximate $w_{Q(T)}(V,\pi)$?

$$\boxed{\begin{array}{c}\text{Non-local games}\\ w_{Q(T)}(V,\pi)\end{array}}$$

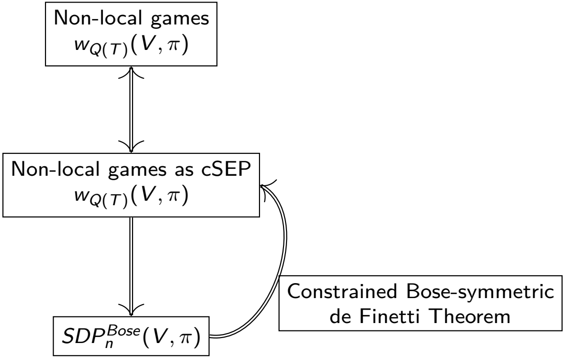
Overview

$$\begin{array}{c} \boxed{\begin{array}{c}\text{Non-local games} \\ w_{Q(T)}(V,\pi)\end{array}} \\ \Big\Updownarrow \\ \boxed{\begin{array}{c}\text{Non-local games as cSEP} \\ w_{Q(T)}(V,\pi)\end{array}} \end{array}$$

$$\begin{tikzcd}
\boxed{\begin{array}{c}\text{Non-local games}\\ w_{Q(T)}(V,\pi)\end{array}} \arrow[d, Leftrightarrow] \\
\boxed{\begin{array}{c}\text{Non-local games as cSEP}\\ w_{Q(T)}(V,\pi)\end{array}} \arrow[d, Rightarrow] \\
\boxed{SDP_n^{Bose}(V,\pi)}
\end{tikzcd}$$

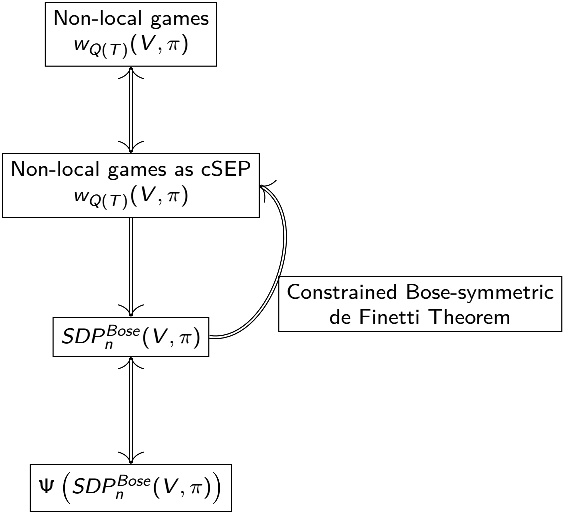
RWTH AACHEN UNIVERSITY

Overview



Non-local games
$w_{Q(T)}(V, \pi)$
Non-local games as cSEP
$w_{Q(T)}(V, \pi)$
$SDP_n^{Bose}(V, \pi)$
Constrained Bose-symmetric de Finetti Theorem

Overview


Non-local games
$w_{Q(T)}(V, \pi)$
Non-local games as cSEP
$w_{Q(T)}(V, \pi)$
$SDP_n^{Bose}(V, \pi)$
Constrained Bose-symmetric de Finetti Theorem
$\Psi\left(SDP_n^{Bose}(V, \pi)\right)$

Non-local Games as Constrained Separability (cSEP) Problems

Overview

$$\begin{tikzcd}
\boxed{\begin{array}{c}\text{Non-local games}\\ w_{Q(T)}(V,\pi)\end{array}} \arrow[d, Leftrightarrow] & \\
\boxed{\begin{array}{c}\text{Non-local games as cSEP}\\ w_{Q(T)}(V,\pi)\end{array}} \arrow[d, Rightarrow] & \\
\boxed{SDP_n^{Bose}(V,\pi)} \arrow[u, Rightarrow, bend right, "\boxed{\begin{array}{c}\text{Constrained Bose-symmetric}\\ \text{de Finetti Theorem}\end{array}}"'] \arrow[d, Leftrightarrow] & \\
\boxed{\Psi\left(SDP_n^{Bose}(V,\pi)\right)} &
\end{tikzcd}$$

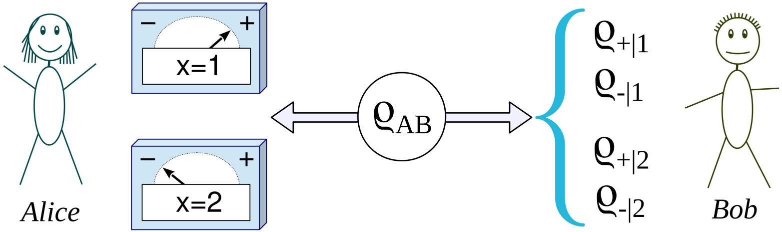


Figure: Schematic description of the steering phenomenon: A state ρ_{AB} is distributed between two parties. Alice performs a measurement (labeled by $x \in \{1, 2\}$) on her particle and obtains the result $a = \pm$. Bob receives the corresponding unnormalized conditional states $\rho_{a|x}$[**Uola˙2020**].

- **Steering:** $\alpha_{\hat{T}}(a|x) := \mathrm{Tr}_T \left[\left(\mathcal{I}_{\hat{T}} \otimes E_T(a|x) \right) \rho_{T\hat{T}} \right]$

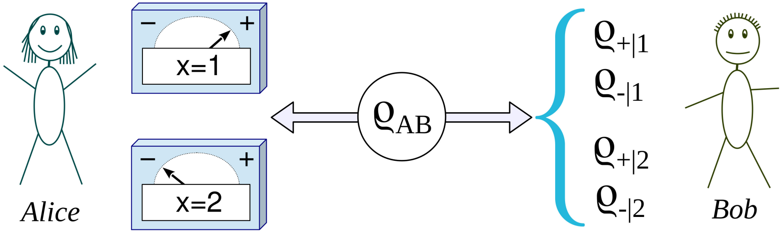


Figure: Schematic description of the steering phenomenon: A state ρ_{AB} is distributed between two parties. Alice performs a measurement (labeled by $x \in \{1, 2\}$) on her particle and obtains the result $a = \pm$. Bob receives the corresponding unnormalized conditional states $\rho_{a|x}$**[Uola˙2020]**.

- **Steering:** $\alpha_{\hat{T}}(a|x) := \mathrm{Tr}_T \left[\left(\mathcal{I}_{\hat{T}} \otimes E_T(a|x) \right) \rho_{T\hat{T}} \right]$
- $+$ some algebraic manipulations
- Before: **Tripartite optimization [jee2020quasi]**
- Now: Bipartite formulation yields **stronger approximation** results in terms of the dimension of quantum assistance

$$w_{Q(T)}(V,\pi) = |T| \max_{(\alpha,D)} \quad \mathrm{Tr}\left[\left(V_{\mathcal{A}\mathcal{B}\mathcal{X}\mathcal{Y}} \otimes S_{\tilde{T}\hat{T}}\right)\left(\alpha_{\mathcal{A}\mathcal{X}\tilde{T}} \otimes D_{\mathcal{B}\mathcal{Y}\hat{T}}\right)\right]$$

$$\text{s.t.} \quad \alpha_{A_1 Q_1 \tilde{T}} = \sum_{a,x} \pi_{\mathcal{X}}(x) \left|ax\right\rangle\left\langle ax\right|_{\mathcal{A}\mathcal{X}} \otimes \alpha_{\tilde{T}}(a|x) \succcurlyeq 0,$$

$$\alpha_{\mathcal{X}\tilde{T}} = \sum_{x} \pi_{\mathcal{X}}(x) \left|x\right\rangle\left\langle x\right|_{\mathcal{X}} \otimes \alpha_{\tilde{T}}, \qquad \mathrm{Tr}\left[\alpha_{\mathcal{A}\mathcal{X}\tilde{T}}\right] = 1$$

$$D_{\mathcal{B}\mathcal{Y}\hat{T}} = \sum_{b,y} \pi_{\mathcal{Y}}(y) \left|by\right\rangle\left\langle by\right|_{\mathcal{B}\mathcal{Y}} \otimes \frac{D_{\hat{T}}(b|y)}{|T|} \succcurlyeq 0,$$

$$D_{\mathcal{Y}\hat{T}} = \sum_{y} \pi_{\mathcal{Y}}(y) \left|y\right\rangle\left\langle y\right|_{\mathcal{Y}} \otimes \frac{\mathcal{I}_{\hat{T}}}{|T|}.$$

$$w_{Q(T)}(V,\pi) = |T| \max_{(\alpha,D)} \quad \mathrm{Tr}\left[\left(V_{\mathcal{ABXY}} \otimes S_{\tilde{T}\hat{T}}\right)\left(\alpha_{\mathcal{AX}\tilde{T}} \otimes D_{\mathcal{BY}\hat{T}}\right)\right]$$

$$\text{s.t.} \quad \alpha_{A_1 Q_1 \tilde{T}} = \sum_{a,x} \pi_{\mathcal{X}}(x) \left|ax\right\rangle \left\langle ax\right|_{\mathcal{AX}} \otimes \alpha_{\tilde{T}}(a|x) \succcurlyeq 0,$$

$$\alpha_{\mathcal{X}\tilde{T}} = \sum_{x} \pi_{\mathcal{X}}(x) \left|x\right\rangle \left\langle x\right|_{\mathcal{X}} \otimes \alpha_{\tilde{T}}, \qquad \mathrm{Tr}\left[\alpha_{\mathcal{AX}\tilde{T}}\right] = 1$$

$$D_{\mathcal{BY}\hat{T}} = \sum_{b,y} \pi_{\mathcal{Y}}(y) \left|by\right\rangle \left\langle by\right|_{\mathcal{BY}} \otimes \frac{D_{\hat{T}}(b|y)}{|T|} \succcurlyeq 0,$$

$$D_{\mathcal{Y}\hat{T}} = \sum_{y} \pi_{\mathcal{Y}}(y) \left|y\right\rangle \left\langle y\right|_{\mathcal{Y}} \otimes \frac{\mathcal{I}_{\hat{T}}}{|T|}.$$

$$\boxed{\text{Constrained product states}} \Longleftrightarrow \boxed{\begin{array}{c}\text{cSEP: convex mixtures}\\ \text{of constrained product states}\\ \Rightarrow \forall x \in \mathcal{X}\end{array}}$$

SEP vs. cSEP

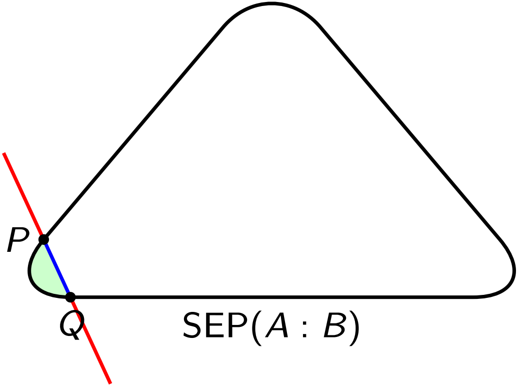


Figure: The convex set of separable states intersected with a constraint hyperplane (or halfspace). P and Q indicate optimal product states satisfying the constraints, while The blue line segment corresponds to the feasible set (convex mixtures of P and Q) of cSEP

.

A broader class of cSEP

$$cSEP(G) := \max_{\rho_{AB} \in \mathcal{B}(AB)} \mathrm{Tr}[G_{AB}(\rho_{AB})]$$

s.t. $\rho_{AB} = \sum_{x \in \mathcal{X}} p(x) \rho_A^x \otimes \rho_B^x$ s.th. $\forall x \in \mathcal{X}$:

$$\rho_A^x \succeq 0,\ \rho_B^x \succeq 0, \qquad \mathrm{Tr}\left[\rho_A^x\right] = 1,\ \mathrm{Tr}\left[\rho_B^x\right] = 1,$$

Local constraints $\forall x \in \mathcal{X}$:

$$\Theta_{A_L \to C_{A_L}}\left(\rho_A^x\right) = W_{C_{A_L}} \otimes \rho_{A_R}^x, \qquad \Upsilon_{B_L \to C_{B_L}}\left(\rho_B^x\right) = K_{C_{B_L}} \otimes \rho_{B_R}^x,$$

Fixed point constraints $\forall x \in \mathcal{X}$:

$$\Omega_{A \to A}\left(\rho_A^x\right) = \rho_A^x, \qquad \Xi_{B \to B}\left(\rho_B^x\right) = \rho_B^x,$$

RWTH AACHEN UNIVERSITY

A broader class of cSEP

$$cSEP(G) := \max_{\rho_{AB} \in \mathcal{B}(AB)} \mathrm{Tr}[G_{AB}(\rho_{AB})]$$

s.t. $\rho_{AB} = \sum_{x \in \mathcal{X}} p(x) \rho_A^x \otimes \rho_B^x$ s.th. $\forall x \in \mathcal{X}$:

$$\rho_A^x \succeq 0,\ \rho_B^x \succeq 0, \qquad \mathrm{Tr}\left[\rho_A^x\right] = 1,\ \mathrm{Tr}\left[\rho_B^x\right] = 1,$$

Local constraints $\forall x \in \mathcal{X}$:

$$\Theta_{A_L \to C_{A_L}}\left(\rho_A^x\right) = W_{C_{A_L}} \otimes \rho_{A_R}^x, \qquad \Upsilon_{B_L \to C_{B_L}}\left(\rho_B^x\right) = K_{C_{B_L}} \otimes \rho_{B_R}^x,$$

Fixed point constraints $\forall x \in \mathcal{X}$:

$$\Omega_{A \to A}\left(\rho_A^x\right) = \rho_A^x, \qquad \Xi_{B \to B}\left(\rho_B^x\right) = \rho_B^x,$$

- **Before:** No fixed point constraints, A_R and B_R one dimensional [**berta2021semidefinite**; **jee2020quasi**]

RWTH AACHEN UNIVERSITY

A broader class of cSEP

$$cSEP(G) := \max_{\rho_{AB} \in \mathcal{B}(AB)} \mathrm{Tr}[G_{AB}(\rho_{AB})]$$

s.t. $\rho_{AB} = \sum_{x \in \mathcal{X}} p(x) \rho_A^x \otimes \rho_B^x$ s.th. $\forall x \in \mathcal{X}$:

$$\rho_A^x \succeq 0,\ \rho_B^x \succeq 0, \qquad \mathrm{Tr}\left[\rho_A^x\right] = 1,\ \mathrm{Tr}\left[\rho_B^x\right] = 1,$$

Local constraints $\forall x \in \mathcal{X}$:

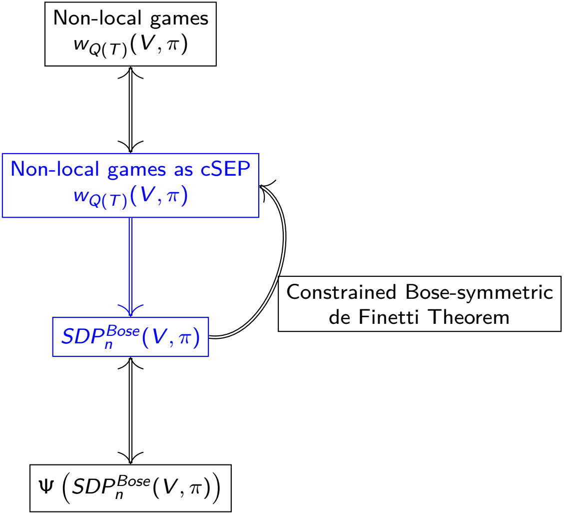
$$\Theta_{A_L \to C_{A_L}}\left(\rho_A^x\right) = W_{C_{A_L}} \otimes \rho_{A_R}^x, \qquad \Upsilon_{B_L \to C_{B_L}}\left(\rho_B^x\right) = K_{C_{B_L}} \otimes \rho_{B_R}^x,$$

Fixed point constraints $\forall x \in \mathcal{X}$:

$$\Omega_{A \to A}\left(\rho_A^x\right) = \rho_A^x, \qquad \Xi_{B \to B}\left(\rho_B^x\right) = \rho_B^x,$$

- **Before:** No fixed point constraints, A_R and B_R one dimensional [**berta2021semidefinite**; **jee2020quasi**]
- SEP problems are already hard and non-linear [**ioannou2007computationalcomplexityquantumseparability**]

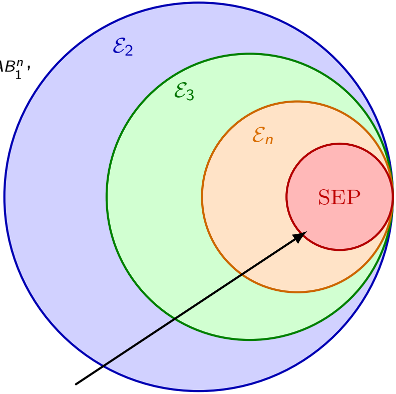
SDP Approximation Hierarchy


Non-local games
$w_{Q(T)}(V, \pi)$
Non-local games as cSEP
$w_{Q(T)}(V, \pi)$
$SDP_n^{Bose}(V, \pi)$
Constrained Bose-symmetric
de Finetti Theorem
$\Psi\left(SDP_n^{Bose}(V, \pi)\right)$

Separability via exchange symmetry

$$\mathcal{D}(A\!:\!B) \supseteq \mathcal{E}_2 \supseteq \mathcal{E}_3 \supseteq \cdots \supseteq \mathcal{E}_n \supseteq \mathrm{SEP}(A\!:\!B)$$

$n-$extendable states:
$\rho_{AB} \in \mathcal{E}_n$:
$\exists \rho_{AB_1^n}$ s.th.: $\forall \pi \in S_n$
$U_{B_1^n}(\pi)\rho_{AB_1^n}U^\dagger_{B_1^n}(\pi) = \rho_{AB_1^n}$,
$\mathrm{tr}_{B_2^n}(\rho_{AB_1^n}) = \rho_{AB}$



Outer approximations:
Converge $n \to \infty$ via
(infinite) de Finetti representation theorems
finite de Finetti yields bounds
on distance to *SEP*

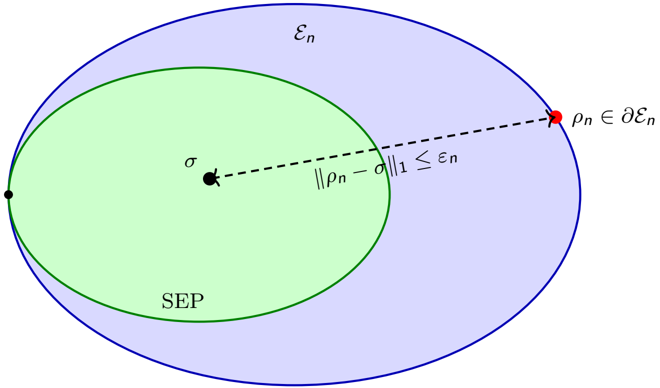


Figure: Finite de Finetti representation theorems bound the distance from n-extendable states to the set of separable states.

cSEP via exchange symmetry

- **Extends to cSEP**, if one carefully treats the constraints (lift to extensions)
- Convergence via adapted finite de Finetti theorems with **information-theoretic** proof technique (analytical bounds on monogamy of entanglement effect) [**berta2021semidefinite**]
- Symmetry admits SDP formulation, but **computationally costly**
- Common approach in unconstrained setting: **SDP symmetry reduction**

cSEP via exchange symmetry

- **Extends to cSEP**, if one carefully treats the constraints (lift to extensions)
- Convergence via adapted finite de Finetti theorems with **information-theoretic** proof technique (analytical bounds on monogamy of entanglement effect) [**berta2021semidefinite**]
- Symmetry admits SDP formulation, but **computationally costly**
- Common approach in unconstrained setting: **SDP symmetry reduction**

Question

*Do symmetry-reduction technique such as **symmetric subspace methods** extend to the constraint setting?*

Definition (Bose-symmetry)

A state $\rho_{B_1^n}$ is Bose-symmetric, if

$$U_{B_1^n}(\pi)\rho_{B_1^n} = \rho_{B_1^n}, \, \forall \pi \in \mathcal{S}_n. \tag{1}$$

Moreover, the state $\rho_{AB_1^n}$ is Bose-symmetric w.r.t. A, if

$$\left(\mathcal{I}_A \otimes U_{B_1^n}(\pi)\right) \rho_{AB_1^n} = \rho_{AB_1^n}, \, \forall \pi \in \mathcal{S}_n. \tag{2}$$

Definition (Bose-symmetry)

A state $\rho_{B_1^n}$ is Bose-symmetric, if

$$U_{B_1^n}(\pi)\rho_{B_1^n} = \rho_{B_1^n},\ \forall \pi \in \mathcal{S}_n. \tag{1}$$

Moreover, the state $\rho_{AB_1^n}$ is Bose-symmetric w.r.t. A, if

$$\left(\mathcal{I}_A \otimes U_{B_1^n}(\pi)\right)\rho_{AB_1^n} = \rho_{AB_1^n},\ \forall \pi \in \mathcal{S}_n. \tag{2}$$

- Stronger Notion of symmetry: $\mathcal{I}_{B_1^n}$ is symmetric, but not Bose-symmetric
- Invariance Eingenspace vs. Invariance Eigenvectors
- No difference for pure states (rank one)!
- $\vee^n(\mathcal{H}) := \left\{ |\psi\rangle \in \mathcal{H}^{\otimes n} \,:\, U_{\mathcal{H}_1^n}(\pi)\,|\psi\rangle = |\psi\rangle\,,\ \forall \pi \in \mathcal{S}_n \right\}$
- One-and-a-half de Finetti [**christandl2007one**], but **no constraints!**
- $\mathrm{End}_{\mathbb{C}[S_n]}\left(\mathcal{H}^{\otimes n}\right) \simeq \vee^n\left(\mathrm{End}\left(\mathcal{H}^{\otimes n}\right)\right)$ vs. $\mathrm{End}_{\mathbb{C}[S_n]}\left(\vee^n(\mathcal{H})\right) \simeq \mathrm{End}\left(\vee^n(\mathcal{H})\right)$

Bose-symmetric SDP hierarchy for cSEP

Lemma

Consider cSEP(G). Then,

$$SDP_n^{Bose}(G) = \max_{\rho_{A(B\bar{B})_1^n}} \mathrm{Tr}\left[G_{AB}\,\rho_{AB}\right]$$

$$\text{s.t.} \quad \rho_{A(B\bar{B})_1^n} \succeq 0, \quad \mathrm{Tr}\left[\rho_{A(B\bar{B})_1^n}\right] = 1, \quad \rho_{A(B\bar{B})_1^n} \textit{ is Bose-symmetric w.r.t. } A,$$

$$\begin{aligned}
&\Theta_{A_L \to C_{A_L}}\left(\rho_{A(B\bar{B})_1^n}\right) = W_{C_{A_L}} \otimes \rho_{A_R(B\bar{B})_1^n}, \\
&\Omega_{A \to A}\left(\rho_{A(B\bar{B})_1^n}\right) = \rho_{A(B\bar{B})_1^n}, \\
&\Upsilon_{(B_L)_n \to C_{B_L}} \circ \mathrm{Tr}_{\bar{B}_n}\left(\rho_{(B\bar{B})_1^n}\right) = K_{C_{B_L}} \otimes \rho_{(B\bar{B})_1^{n-1}(B_R)_n}, \\
&\Xi_{B_n \to B_n} \circ \mathrm{Tr}_{\bar{B}_n}\left(\rho_{(B\bar{B})_1^n}\right) = \rho_{(B\bar{B})_1^{n-1}B_n},
\end{aligned} \tag{3}$$

such that

$$0 \leq SDP_n^{Bose}(G) - cSEP(G) \leq \min\left\{f(A|B\bar{B}), f(B\bar{B}|\cdot)\right\}\sqrt{4\ln(2)}\sqrt{\frac{\ln(d_A)}{n}}. \tag{4}$$

AACHEN
VERSITY

$w_{Q(T)}(V, \pi) \Rightarrow SDP_n^{Bose}(T, V, \pi)$ (or $cSEP(G) \Rightarrow SDP_n^{Bose}(G)$)

- Separable states as convex mixture of pure product state? **No**, additional linear constraints...

$w_{Q(T)}(V, \pi) \Rightarrow SDP_n^{Bose}(T, V, \pi)$ (or $cSEP(G) \Rightarrow SDP_n^{Bose}(G)$)

- Separable states as convex mixture of pure product state? **No**, additional linear constraints...
- **Local purification scheme** $\rightarrow$ embedding of permutation-invariant states subject to linear constraints into the symmetric subspace of a higher-dimensional space, intersected with the corresponding constraints.

$w_{Q(T)}(V,\pi) \Rightarrow SDP_n^{Bose}(T,V,\pi)$ (or $cSEP(G) \Rightarrow SDP_n^{Bose}(G)$)

- Separable states as convex mixture of pure product state? **No**, additional linear constraints...
- **Local purification scheme** $\rightarrow$ embedding of permutation-invariant states subject to linear constraints into the symmetric subspace of a higher-dimensional space, intersected with the corresponding constraints.
-

$$\rho_{A(B\bar{B})_1^n} := \sum_{x\in\mathcal{X}} p(x)\,\rho_A^x \otimes \left(|\psi^x\rangle\langle\psi^x|_{B\bar{B}}\right)^{\otimes n}, \quad |\psi^x\rangle_{B\bar{B}} := \left(\sqrt{\rho_B^x}\otimes\mathcal{I}_{\bar{B}}\right)|\Psi\rangle_{B\bar{B}}$$

- Separable states as convex mixture of pure product state? **No**, additional linear constraints...
- **Local purification scheme** $\rightarrow$ embedding of permutation-invariant states subject to linear constraints into the symmetric subspace of a higher-dimensional space, intersected with the corresponding constraints.
-
$$\rho_{A(B\bar{B})_1^n} := \sum_{x\in\mathcal{X}} p(x)\,\rho_A^x \otimes \left(\left|\psi^x\right\rangle \left\langle\psi^x\right|_{B\bar{B}}\right)^{\otimes n}, \quad \left|\psi^x\right\rangle_{B\bar{B}} := \left(\sqrt{\rho_B^x} \otimes \mathcal{I}_{\bar{B}}\right) \left|\Psi\right\rangle_{B\bar{B}}$$
- Candidate is **Bose-symmetric** and **satisfies the linear constraints** in $SDP_n^{Bose}(T,V,\pi)$

De Finetti Theorems

$$
\begin{tikzcd}
\boxed{\text{Non-local games } w_{Q(T)}(V,\pi)} \arrow[d, Leftrightarrow] \\
\boxed{\text{Non-local games as cSEP } w_{Q(T)}(V,\pi)} \arrow[d, Rightarrow] \\
\boxed{SDP_n^{Bose}(V,\pi)} \arrow[u, Rightarrow, bend right=60, swap, "\boxed{\text{Constrained Bose-symmetric de Finetti Theorem}}"] \arrow[d, Leftrightarrow] \\
\boxed{\Psi\left(SDP_n^{Bose}(V,\pi)\right)}
\end{tikzcd}
$$

Lemma: Constrained Bose-symmetric de Finetti Theorem

Let $\rho_{A(B\bar{B})_1^n}$ be Bose-symmetric with respect to A. Furthermore, consider linear mappings $\Theta_{A_L \to C_{A_L}}$, $\Omega_{A\to A}$, $\Upsilon_{B_L \to C_{B_L}}$, $\Xi_{B\to B}$ and operators $W_{C_{A_L}}$, $K_{C_{B_L}}$. Assuming

$$\Theta_{A_L \to C_{A_L}}\left(\rho_{A(B\bar{B})_1^n}\right) = W_{C_{A_L}} \otimes \rho_{A_R(B\bar{B})_1^n}, \qquad \Omega_{A\to A}\left(\rho_{A(B\bar{B})_1^n}\right) = \rho_{A(B\bar{B})_1^n},$$
$$\Upsilon_{(B_L)_n \to C_{B_L}} \circ \mathrm{Tr}_{\bar{B}_n}\left(\rho_{(B\bar{B})_1^n}\right) = K_{C_{B_L}} \otimes \rho_{(B\bar{B})_1^{n-1}(B_R)_n}, \qquad \Xi_{B_n \to B_n} \circ \mathrm{Tr}_{\bar{B}_n}\left(\rho_{(B\bar{B})_1^n}\right) = \rho_{(B\bar{B})_1^{n-1}B_n},$$

then there exists a probability distribution $\{p(x)\}_{x\in\mathcal{X}}$ and states $\{\sigma_A^x\}_{x\in\mathcal{X}}$, $\left\{\sigma_{B\bar{B}}^x\right\}_{x\in\mathcal{X}}$ such that

$$\left\|\rho_{A(B\bar{B})} - \sum_{x\in\mathcal{X}} p(x)\sigma_A^x \otimes \sigma_{(B\bar{B})}^x\right\|_1 \leq \min\left\{f(A|B\bar{B}), f(B\bar{B}|\cdot)\right\}\sqrt{4\ln(2)}\sqrt{\frac{\ln(|A|)}{n}}$$

where $f(A|B\bar{B}), f(B\bar{B}|\cdot)$ are minimal measurement distortions and for each $x \in \mathcal{X}$

$$\Theta_{A_L \to C_{A_L}}(\sigma_A^x) = W_{C_{A_L}} \otimes \sigma_{A_R}^x, \qquad \Omega_{A\to A}(\sigma_A^x) = \sigma_A^x,$$
$$\Upsilon_{B_L \to C_{B_L}} \circ \mathrm{Tr}_{\bar{B}}\left(\sigma_{(B\bar{B})}^x\right) = K_{C_{B_L}} \otimes \sigma_{B_R}^x, \qquad \Xi_{B\to B_n} \circ \mathrm{Tr}_{\bar{B}}\left(\sigma_{(B\bar{B})}^x\right) = \sigma_B^x.$$

RWTH AACHEN UNIVERSITY

Lemma: Constrained Bose-symmetric de Finetti Theorem

Let $\rho_{A(B\bar{B})_1^n}$ be Bose-symmetric with respect to A. Furthermore, consider linear mappings $\Theta_{A_L\to C_{A_L}}$, $\Omega_{A\to A}$, $\Upsilon_{B_L\to C_{B_L}}$, $\Xi_{B\to B}$ and operators $W_{C_{A_L}}$, $K_{C_{B_L}}$. Assuming

$$\Theta_{A_L\to C_{A_L}}\left(\rho_{A(B\bar{B})_1^n}\right) = W_{C_{A_L}} \otimes \rho_{A_R(B\bar{B})_1^n}, \qquad \Omega_{A\to A}\left(\rho_{A(B\bar{B})_1^n}\right) = \rho_{A(B\bar{B})_1^n},$$

$$\Upsilon_{(B_L)_n\to C_{B_L}} \circ \mathrm{Tr}_{\bar{B}_n}\left(\rho_{(B\bar{B})_1^n}\right) = K_{C_{B_L}} \otimes \rho_{(B\bar{B})_1^{n-1}(B_R)_n}, \qquad \Xi_{B_n\to B_n} \circ \mathrm{Tr}_{\bar{B}_n}\left(\rho_{(B\bar{B})_1^n}\right) = \rho_{(B\bar{B})_1^{n-1}B_n},$$

then there exists a probability distribution $\{p(x)\}_{x\in\mathcal{X}}$ and states $\{\sigma_A^x\}_{x\in\mathcal{X}}$, $\left\{\sigma_{B\bar{B}}^x\right\}_{x\in\mathcal{X}}$ such that

$$\left\|\rho_{A(B\bar{B})} - \sum_{x\in\mathcal{X}} p(x)\sigma_A^x \otimes \sigma_{(B\bar{B})}^x\right\|_1 \leq \min\left\{f(A|B\bar{B}), f(B\bar{B}|\cdot)\right\}\sqrt{4\ln(2)}\sqrt{\frac{\ln(|A|)}{n}}$$

where $f(A|B\bar{B}), f(B\bar{B}|\cdot)$ are minimal measurement distortions and for each $x\in\mathcal{X}$

$$\Theta_{A_L\to C_{A_L}}(\sigma_A^x) = W_{C_{A_L}} \otimes \sigma_{A_R}^x, \qquad \Omega_{A\to A}(\sigma_A^x) = \sigma_A^x,$$

$$\Upsilon_{B_L\to C_{B_L}} \circ \mathrm{Tr}_{\bar{B}}\left(\sigma_{(B\bar{B})}^x\right) = K_{C_{B_L}} \otimes \sigma_{B_R}^x, \qquad \Xi_{B\to B_n} \circ \mathrm{Tr}_{\bar{B}}\left(\sigma_{(B\bar{B})}^x\right) = \sigma_B^x.$$

- **ALSO:** New constrained de Finetti Theorem for **symmetric states** w/ application to new cSEP

Sketch of Proof I

- Conditioning with classical-quantum (cq) states

$$\rho_{A|z} = \frac{\mathrm{Tr}_Z\left[\left(\mathcal{I}_A \otimes M^z_{B\to Z}\right)\rho_{AB}\right]}{\mathrm{Tr}\left[\left(\mathcal{I}_A \otimes M^z_B\right)\rho_{AB}\right]} \tag{5}$$

Sketch of Proof I

- Conditioning with classical-quantum (cq) states

$$\rho_{A|z} = \frac{\mathrm{Tr}_Z\left[\left(\mathcal{I}_A \otimes M^z_{B\to Z}\right)\rho_{AB}\right]}{\mathrm{Tr}\left[\left(\mathcal{I}_A \otimes M^z_B\right)\rho_{AB}\right]} \tag{5}$$

- The Self-decoupling Lemma [**brandao2013product**]: an information theoretic proof technique for symmetric cq-states

$$\mathbb{E}_{z_1^m}\left\|\rho_{AZ_{m+1}|z_1^m} - \rho_{A|z_1^m}\otimes\rho_{Z_{m+1}|z_1^m}\right\|_1 \leq \frac{2\ln 2\log|A|}{n}. \tag{6}$$

Sketch of Proof I

- Conditioning with classical-quantum (cq) states

$$\rho_{A|z} = \frac{\mathrm{Tr}_Z \left[\left(\mathcal{I}_A \otimes M^z_{B\to Z} \right) \rho_{AB} \right]}{\mathrm{Tr} \left[\left(\mathcal{I}_A \otimes M^z_B \right) \rho_{AB} \right]} \tag{5}$$

- The Self-decoupling Lemma [**brandao2013product**]: an information theoretic proof technique for symmetric cq-states

$$\mathbb{E}_{z_1^m} \left\| \rho_{AZ_{m+1}|z_1^m} - \rho_{A|z_1^m} \otimes \rho_{Z_{m+1}|z_1^m} \right\|_1 \leq \frac{2 \ln 2 \log |A|}{n}. \tag{6}$$

- Replace classical-quantum states and use convexity

$$\left\| \rho_{A\left(B\bar{B}\right)_{m+1}} - \sum_{z_1^m} p(z_1^m)\, \rho_{A|z_1^m} \otimes \rho_{\left(B\bar{B}\right)_{m+1}|z_1^m} \right\|_1 \leq g\left(|A|, |B|\right) \sqrt{\frac{2 \ln 2 \log |A|}{n}}. \tag{7}$$

Sketch of Proof I

- Conditioning with classical-quantum (cq) states

$$\rho_{A|z} = \frac{\mathrm{Tr}_Z \left[\left(\mathcal{I}_A \otimes M^z_{B\to Z} \right) \rho_{AB} \right]}{\mathrm{Tr} \left[\left(\mathcal{I}_A \otimes M^z_B \right) \rho_{AB} \right]} \tag{5}$$

- The Self-decoupling Lemma [**brandao2013product**]: an information theoretic proof technique for symmetric cq-states

$$\mathbb{E}_{z_1^m} \left\| \rho_{AZ_{m+1}|z_1^m} - \rho_{A|z_1^m} \otimes \rho_{Z_{m+1}|z_1^m} \right\|_1 \leq \frac{2 \ln 2 \log |A|}{n}. \tag{6}$$

- Replace classical-quantum states and use convexity

$$\left\| \rho_{A(B\bar{B})_{m+1}} - \sum_{z_1^m} p(z_1^m)\, \rho_{A|z_1^m} \otimes \rho_{(B\bar{B})_{m+1}|z_1^m} \right\|_1 \leq g\left(|A|, |B|\right) \sqrt{\frac{2 \ln 2 \log |A|}{n}}. \tag{7}$$

- $g\left(|A|, |B|\right) = \min\left(f(A|B),\ f(B|)\cdot\right)$ can be bounded for informationally complete measurements

Sketch of Proof I

- Conditioning with classical-quantum (cq) states

$$\rho_{A|z} = \frac{\mathrm{Tr}_Z\left[\left(\mathcal{I}_A \otimes M^z_{B\to Z}\right)\rho_{AB}\right]}{\mathrm{Tr}\left[\left(\mathcal{I}_A \otimes M^z_{B}\right)\rho_{AB}\right]} \tag{5}$$

- The Self-decoupling Lemma [**brandao2013product**]: an information theoretic proof technique for symmetric cq-states

$$\mathbb{E}_{z_1^m}\left\|\rho_{AZ_{m+1}|z_1^m} - \rho_{A|z_1^m}\otimes\rho_{Z_{m+1}|z_1^m}\right\|_1 \leq \frac{2\ln 2\log|A|}{n}. \tag{6}$$

- Replace classical-quantum states and use convexity

$$\left\|\rho_{A\left(B\bar{B}\right)_{m+1}} - \sum_{z_1^m} p\left(z_1^m\right)\rho_{A|z_1^m}\otimes\rho_{\left(B\bar{B}\right)_{m+1}|z_1^m}\right\|_1 \leq g\left(|A|,|B|\right)\sqrt{\frac{2\ln 2\log|A|}{n}}. \tag{7}$$

- $g\left(|A|,|B|\right) = \min\left(f(A|B),\ f(B|)\cdot\right)$ can be bounded for informationally complete measurements
- Standard de Finetti representation theorems: No guarantee that the measure on states not satisfying the constraints is approximately zero

Sketch of Proof II

- Bose-symmetry $\Rightarrow$ symmetry $\Rightarrow$ symmetry of marginals

Sketch of Proof II

- Bose-symmetry ⇒ symmetry ⇒ symmetry of marginals
- **Symmetry of linear maps** from symmetry of state
- Linear maps **commuting with measurements**

Sketch of Proof II

- Bose-symmetry $\Rightarrow$ symmetry $\Rightarrow$ symmetry of marginals
- **Symmetry of linear maps** from symmetry of state
- Linear maps **commuting with measurements**
- Recall: $\Upsilon_{(B_L)_n \to C_{B_L}} \circ \mathrm{Tr}_{\bar{B}_n}\left(\rho_{(B\bar{B})_1^n}\right) = K_{C_{B_L}} \otimes \rho_{(B\bar{B})_1^{n-1}(B_R)_n}$

Sketch of Proof II

- Bose-symmetry $\Rightarrow$ symmetry $\Rightarrow$ symmetry of marginals
- **Symmetry of linear maps** from symmetry of state
- Linear maps **commuting with measurements**
- Recall: $\Upsilon_{(B_L)_n \to C_{B_L}} \circ \mathrm{Tr}_{\bar{B}_n}\left(\rho_{(B\bar{B})_1^n}\right) = K_{C_{B_L}} \otimes \rho_{(B\bar{B})_1^{n-1}(B_R)_n}$

$$
\begin{aligned}
&\Upsilon_{(B_L)_{m+1} \to C_{B_L}} \circ \mathrm{Tr}_{(\bar{B})_{m+1}}\left[\rho_{(B\bar{B})_{m+1}|z_1^m}\right] \\
&= \frac{1}{N}\,\mathrm{Tr}_{Z_1^m (B\bar{B})_{m+2}^n}\left[\left(M^{z_1^m}_{(B\bar{B})_1^m} \otimes \mathcal{I}_{C_{B_L}(B_R)_{m+1}(B\bar{B})_{m+2}^n}\right)\Upsilon_{(B_L)_{m+1} \to C_{B_L}} \circ \mathrm{Tr}_{(\bar{B})_{m+1}}\left[\rho_{(B\bar{B})_1^n}\right]\right] \\
&= \frac{1}{N}\,\mathrm{Tr}_{Z_1^m (B\bar{B})_{m+2}^n}\left[\left(M^{z_1^m}_{(B\bar{B})_1^m} \otimes \mathcal{I}_{C_{B_L}(B_R)_{m+1}(B\bar{B})_{m+2}^n}\right)\rho_{(B\bar{B})_1^m} \otimes K_{C_{B_L}} \otimes \rho_{(B_R)_{m+1}(B\bar{B})_{m+2}^n}\right] \\
&= K_{C_{B_L}} \otimes \rho_{(B_R)_{m+1}|z_1^m}
\end{aligned}
$$

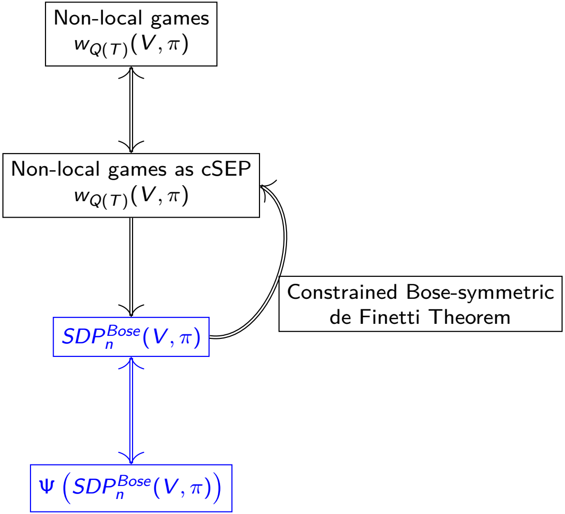
Sketch of Proof II

- Bose-symmetry $\Rightarrow$ symmetry $\Rightarrow$ symmetry of marginals
- **Symmetry of linear maps** from symmetry of state
- Linear maps **commuting with measurements**
- Recall: $\Upsilon_{(B_L)_n \to C_{B_L}} \circ \mathrm{Tr}_{\bar{B}_n}\left(\rho_{(B\bar{B})_1^n}\right) = K_{C_{B_L}} \otimes \rho_{(B\bar{B})_1^{n-1}(B_R)_n}$

$$
\begin{aligned}
&\Upsilon_{(B_L)_{m+1} \to C_{B_L}} \circ \mathrm{Tr}_{(\bar{B})_{m+1}}\left[\rho_{(B\bar{B})_{m+1}|z_1^m}\right] \\
&= \frac{1}{N}\,\mathrm{Tr}_{Z_1^m (B\bar{B})_{m+2}^n}\left[\left(M^{z_1^m}_{(B\bar{B})_1^m} \otimes \mathcal{I}_{C_{B_L}(B_R)_{m+1}(B\bar{B})_{m+2}^n}\right) \Upsilon_{(B_L)_{m+1} \to C_{B_L}} \circ \mathrm{Tr}_{(\bar{B})_{m+1}}\left[\rho_{(B\bar{B})_1^n}\right]\right] \\
&= \frac{1}{N}\,\mathrm{Tr}_{Z_1^m (B\bar{B})_{m+2}^n}\left[\left(M^{z_1^m}_{(B\bar{B})_1^m} \otimes \mathcal{I}_{C_{B_L}(B_R)_{m+1}(B\bar{B})_{m+2}^n}\right) \rho_{(B\bar{B})_1^m} \otimes K_{C_{B_L}} \otimes \rho_{(B_R)_{m+1}(B\bar{B})_{m+2}^n}\right] \\
&= K_{C_{B_L}} \otimes \rho_{(B_R)_{m+1}|z_1^m}
\end{aligned}
$$

In paper: actually do the measurement to obtain inner points $\Rightarrow$ **Convergent inner sequence** via measurement-based rounding scheme (Soon on arXiv: efficient algorithm)

Efficient Approximation Algorithm


Non-local games
$w_{Q(T)}(V,\pi)$
Non-local games as cSEP
$w_{Q(T)}(V,\pi)$
$SDP_n^{Bose}(V,\pi)$
Constrained Bose-symmetric de Finetti Theorem
$\Psi\left(SDP_n^{Bose}(V,\pi)\right)$

Why Bose-symmetry?

Restriction to polynomial in n operator space:

- $\Theta_{A_L \to C_{A_L}}\left(\rho_{A\left(B\bar{B}\right)_1^n}\right),\ W_{C_{A_L}} \otimes \rho_{A_R\left(B\bar{B}\right)_1^n} \in \mathrm{End}_{\mathbb{C}[S_n]}\left(C_{A_L} \otimes A_R \otimes \vee^n\left(B\bar{B}\right)\right)$
- $\Omega_{A \to A}\left(\rho_{A\left(B\bar{B}\right)_1^n}\right),\ \rho_{A\left(B\bar{B}\right)_1^n} \in \mathrm{End}_{\mathbb{C}[S_n]}\left(A \otimes \vee^n\left(B\bar{B}\right)\right)$
- $\Xi_{B_n \to B_n} \circ \mathrm{Tr}_{\bar{B}_n}\left(\rho_{\left(B\bar{B}\right)_1^n}\right),\ \rho_{\left(B\bar{B}\right)_1^{n-1}B_n} \in \mathrm{End}_{\mathbb{C}[S_n]}\left(\vee^{n-1}\left(B\bar{B}\right) \otimes B_n\right)$
- $\Upsilon_{(B_L)_n \to C_{B_L}} \circ \mathrm{Tr}_{\bar{B}_n}\left(\rho_{\left(B\bar{B}\right)_1^n}\right),\ K_{C_{B_L}} \otimes \rho_{\left(B\bar{B}\right)_1^{n-1}(B_R)_n} \in$ $\mathrm{End}_{\mathbb{C}[S_{n-1}]}\left(C_{B_L} \otimes \vee^{n-1}\left(B\bar{B}\right) \otimes (B_R)_n\right)$

Bottleneck

Problem

A priori,

$$SDP_n^{Bose}(G) = \max_{\rho_{A(B\bar{B})_1^n} \in End_{\mathbb{C}[S_n]}\left(A \otimes \vee^n (B\bar{B})\right)} \mathrm{Tr}\left[G_{AB}\,\rho_{AB}\right]$$

$$\begin{aligned} s.t. \quad & \rho_{A(B\bar{B})_1^n} \succeq 0, \quad \mathrm{Tr}\left[\rho_{A(B\bar{B})_1^n}\right] = 1, \\ & \Theta_{A_L \to C_{A_L}}\left(\rho_{A(B\bar{B})_1^n}\right) = W_{C_{A_L}} \otimes \rho_{A_R (B\bar{B})_1^n}, \\ & \Omega_{A \to A}\left(\rho_{A(B\bar{B})_1^n}\right) = \rho_{A(B\bar{B})_1^n}, \\ & \Upsilon_{(B_L)_n \to C_{B_L}} \circ \mathrm{Tr}_{\bar{B}_n}\left(\rho_{(B\bar{B})_1^n}\right) = K_{C_{B_L}} \otimes \rho_{(B\bar{B})_1^{n-1}(B_R)_n}, \\ & \Xi_{B_n \to B_n} \circ \mathrm{Tr}_{\bar{B}_n}\left(\rho_{(B\bar{B})_1^n}\right) = \rho_{(B\bar{B})_1^{n-1} B_n}, \end{aligned}$$

is not given in a symmetry adapted basis. The objects in the canonical basis are of exponential size in n.

Question

Can we construct a symmetry-adapted SDP hierarchy, whose size scales polynomially with n, efficiently with respect to n?

Schur-Weyl duality: matrix decomposition & efficient algorithm

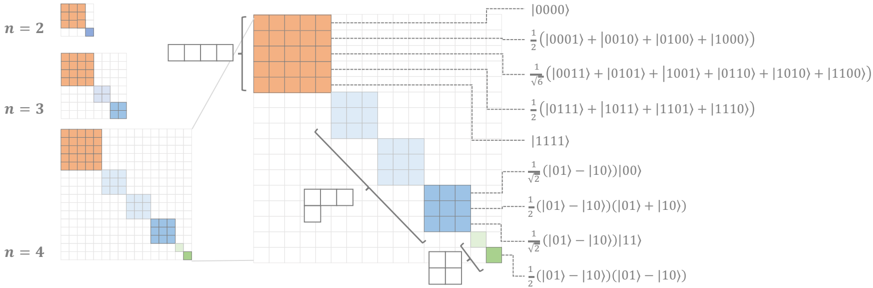


Figure: Matrix decomposition under Schur-Weyl duality. Blocks are labeled by irreducible representations. Bose-symmetric operators correspond to the orange block [**Anschuetz˙2023**].

Benefits Bose-Symmetry: Orthonormal basis, local dimension vs. extensions

Schur-Weyl duality: matrix decomposition & efficient algorithm

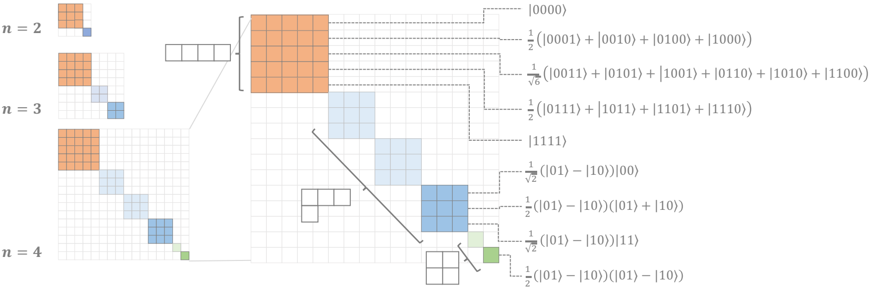


Figure: Matrix decomposition under Schur-Weyl duality. Blocks are labeled by irreducible representations. Bose-symmetric operators correspond to the orange block [**Anschuetz˙2023**].
Benefits Bose-Symmetry: Orthonormal basis, local dimension vs. extensions

Proposition

For fixed local dimension, the transformation into a Schur basis to the symmetric subspace can be computed efficiently. Hence, $\Psi\left(SDP_n^{Bose}(V,\pi)\right)$ *can be constructed efficiently w.r.t. n.*

Polynomial time algorithm

Theorem

For a two-player free non-local game with $|A|$ answers, $|Q|$ questions and quantum assistance of size $|T|$, there exists a poly$\left(\epsilon^{-1}\right)$ costly transformation to an SDP with

$$\left(\epsilon^{-4} \cdot \mathrm{poly}\left(|A|, |Q|, |T|\right)\right)^{(|A||Q||T|)^2},$$

many variables, which approximates the value of the game up to an additive ϵ-error.

Outlook

Question (cSEP)

*The cSEP formalism admits convergent approximation hierarchies derived from de Finetti–type reductions. A natural question is whether this framework can be extended to handle **global constraints**, or whether such constraints force the use of alternative constructions, such as polarization techniques [***Plvala2025***]. Additionally, does one obtain a **certificate of finite convergence**, similar in spirit to the rank-loop criteria appearing in the NPA hierarchy [***pironio2010convergent***]?*

Question (Complexity)

*Do other variants of **dual pairs** such as e.g. the mixed Schur-Weyl duality [***grinko2023gelfandtsetlinbasispartiallytransposed***] or the Schur-Weyl duality for the Clifford group [***gross2021schur***] allow for similar symmetry reductions?*

Thank you for listening!

RWTH AACHEN UNIVERSITY

Convergent inner sequence

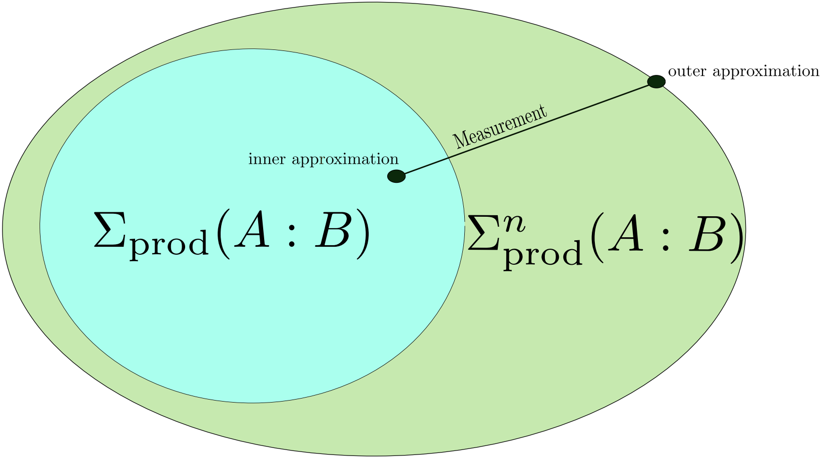


Figure: Rounding scheme

$$SDP_n^{Bose}(T, V, \pi) = |T| \max_{\rho} \mathrm{Tr}\left[\left(V_{A_1 A_2 Q_1 Q_2} \otimes S_{T\tilde{T}}\right) \rho_{(A_1 Q_1 T)(A_2 Q_2 \tilde{T})}\right]$$

s.t.

$$\rho_{(A_1 Q_1 T)(A_2 Q_2 \tilde{T})} = \mathrm{Tr}_{\left((A_2 Q_2 \tilde{T})\left(\overline{A_2 Q_2 \tilde{T}}\right)\right)_2^n}\left[\rho_{(A_1 Q_1 T)\left((A_2 Q_2 \tilde{T})(\overline{A_2 Q_2 \tilde{T}})\right)_1^n}\right]$$

$\rho_{(A_1 Q_1 T)\left((A_2 Q_2 \tilde{T})(\overline{A_2 Q_2 \tilde{T}})\right)_1^n}$ is Bose-symmetric state w.r.t. $(A_1 Q_1 T)$

$$\mathrm{Tr}_{A_1}\left[\rho_{(A_1 Q_1 T)\left((A_2 Q_2 \tilde{T})(\overline{A_2 Q_2 \tilde{T}})\right)_1^n}\right] = \sum_{q_1 \in Q_1} \pi_1(q_1) \left|q_1\right\rangle \left\langle q_1\right|_{Q_1} \otimes \rho_{T\left((A_2 Q_2 \tilde{T})(\overline{A_2 Q_2 \tilde{T}})\right)_1^n}$$

$$\mathrm{Tr}_{(A_2)_n \overline{(A_2 Q_2 \tilde{T})_n}}\left[\rho_{\left((A_2 Q_2 \tilde{T})(\overline{A_2 Q_2 \tilde{T}})\right)_1^n}\right]$$
$$= \rho_{\left((A_2 Q_2 \tilde{T})(\overline{A_2 Q_2 \tilde{T}})\right)_1^{n-1}} \otimes \left(\sum_{q_2 \in Q_2} \pi_2(q_2) \left|q_2\right\rangle \left\langle q_2\right|_{Q_2} \otimes \frac{\mathcal{I}_{\tilde{T}}}{|T|}\right).$$

RWTH AACHEN UNIVERSITY

Schur-Weyl duality

- $\mathrm{End}_{\mathbb{C}[S_n]}\left(\vee^n(\mathcal{H})\right) \simeq \mathbb{C}^{m_{(n)} \times m_{(n)}}$, $m_{(n)} = \binom{n+|\mathcal{H}|-1}{n}$
- A $*$-isomorphism (change of basis) preserving semi-definiteness and the value of the SDP build from semistandard Young tableaux $\mathcal{T}_{\lambda,d}$:

$$\begin{aligned} \psi : \mathrm{End}_{\mathbb{C}[S_n]}\left(\vee^n(\mathcal{H})\right) &\to \mathbb{C}^{m_{(n)} \times m_{(n)}} \\ A &\mapsto \left(\langle A u_\tau, u_\gamma \rangle_{S_n}\right)_{\tau,\gamma \in \mathcal{T}_{\lambda,d}} \end{aligned} \tag{8}$$

with

$$\mathbb{R}^{d_{\mathcal{H}}^n} \ni u_\tau = \sum_{\tau' \sim \tau} \bigotimes_{y \in Y((n))} e_{\tau'(y)} \tag{9}$$

- Example:

$$\tau = \boxed{1 \mid 2 \mid 2} \tag{10}$$

and corresponding Young symmetrizer

$$u_\tau = |011\rangle + |101\rangle + |110\rangle\,. \tag{11}$$

Problem

Can we do this efficiently w.r.t. n?

- Bose-symmetry $\Rightarrow$ **"Redundancies"** in canonical basis representation
- $$\begin{bmatrix} b & & & \\ & a & a & \\ & a & a & \\ & & & c \end{bmatrix}$$
- **Efficient** representation of Bose-symmetric orbitals via contingency tables with fixed margins
- Poly(n)-queries to an oracle

- New (Schur) basis: Bose-symmetric orbital averages

Lemma

Let τ, γ be two semi-standard labellings of $Y\left((n)\right)$. Given complex numbers $\left\{z_{\vec{t},\vec{t'}}\right\}_{\vec{t},\vec{t'}\in\mathcal{T}_{(n),\,d_{\mathcal{H}}}}$, then

$$\sum_{\vec{t},\vec{t'}\in\mathcal{T}_{d_{\mathcal{H}},n}} z_{\vec{t},\vec{t'}}\, u_\tau^T\, C_{\vec{t},\vec{t'}}^{\vee^n} u_\gamma \tag{12}$$

can be computed efficiently.

Proof sketch: Use dual spaces [**polak2020new**]:

$$\begin{aligned}\sum_{\vec{t},\vec{t'}\in\mathcal{T}_{(n),\,d_{\mathcal{H}}}} u_\tau^T\, C_{\vec{t},\vec{t'}}^{\vee^n} u_\gamma \mu\left(\vec{t},\vec{t'}\right) &= \sum_{\vec{t},\vec{t'}\in\mathcal{T}_{(n),\,d_{\mathcal{H}}}} g\left(C_{\vec{t},\vec{t'}}^{\vee^n}\right)\mu\left(\vec{t},\vec{t'}\right)\\ &= \sum_{\vec{t},\vec{t'}\in\mathcal{T}_{(n),\,d_{\mathcal{H}}}} \binom{n}{\vec{t}}\cdot\binom{n}{\vec{t'}}\delta_{\vec{t},\tau}\delta_{\vec{t'},\gamma}\mu\left(\vec{t},\vec{t'}\right) = \binom{n}{\tau}\cdot\binom{n}{\gamma}\mu\left(\tau,\mu\right)\end{aligned}$$

- Never construct exponentially sized $SDP_n^{Bose}(V,\pi)$
- $\Psi\left(SDP_n^{Bose}(V,\pi)\right)$ can be efficiently constructed by:
  - All coefficients of the expressions in $SDP_n^{Bose}(V,\pi)$ are efficiently computable
  - Previous Lemma yields the efficient construction of $\Psi\left(SDP_n^{Bose}(V,\pi)\right)$
- Since $\Psi\left(SDP_n^{Bose}(V,\pi)\right) = SDP_n^{Bose}(V,\pi)$, Bose-symmetric de Finetti bound translates to $\Psi\left(SDP_n^{Bose}(V,\pi)\right)$